THE INTEGRAL CLOSURE OF THE GROUND FIELD IN THE GLOBAL SECTIONS OF A CONNECTED REDUCED SCHEME IS A FIELD

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Allowed standard prerequisites (editorial).

- Integral ring extensions have no inclusion of prime ideals and their spectra are totally disconnected. Stacks Project, Algebra, Lemmas algebra-lemma-integral-no-inclusion (tag 00GS) and algebra-lemma-ring-with-only-minimal-primes (tag 00G3). These give that every prime of the integral closure is minimal and maximal, which is what makes the constancy argument work.
- Morphisms into affine schemes and the characterization of local rings. Stacks Project, Schemes, Lemma schemes-lemma-morphism-into-affine (tag 01I1); Algebra, Lemma algebra-lemma-characterize-local-ring (tag 00ER). The first factors the structure morphism through Spec(k') and the second identifies k' as local once all elements outside p are units.

Source reference. Original reference 01I1: sources/stacks-chow/tags.txt, source lines 1962–1962. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00GT: sources/stacks-chow/tags.txt, source lines 619–619. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04MG: sources/stacks-chow/tags.txt, source lines 6042–6042. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00E9: sources/stacks-chow/tags.txt, source lines 526–526. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00EU: sources/stacks-chow/tags.txt, source lines 548–548. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a scheme over k which is connected and reduced. Then the integral closure of k in $\Gamma(X, \mathcal{O}_X)$ is a field.*

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Proof. Let $k' \subset \Gamma(X, \mathcal{O}_X)$ be the integral closure of k. Then $X \to \mathrm{Spec}(k)$ factors through $\mathrm{Spec}(k')$, see Schemes, Lemma 01I1. As X is reduced we see that k' has no nonzero nilpotent elements. As $k \to k'$ is integral we see that every prime ideal of k' is both a maximal ideal and a minimal prime, and $\mathrm{Spec}(k')$ is totally disconnected, see Algebra, Lemmas 00GT and 04MG. As X is connected the morphism $X \to \mathrm{Spec}(k')$ is constant, say with image the point corresponding to $\mathfrak{p} \subset k'$. Then any $f \in k'$, $f \notin \mathfrak{p}$ maps to an invertible element of $\mathcal{O}_X$. By definition of k' this then forces f to be a unit of k'. Hence we see that k' is local with maximal ideal $\mathfrak{p}$, see Algebra, Lemma 00E9. Since we've already seen that k' is reduced this implies that k' is a field, see Algebra, Lemma 00EU. $\square$